
Stop Overcomplicating Your Vector DB

1 Why Postgres is all you need for 90% of AI apps

2 Cut complexity and infrastructure costs

The Standalone DB Trap

- 1 Separate vector DBs cause painful data sync issues.
- 2 High monthly cloud bills for small datasets (<50M vectors).
- 3 Dual-system transactions increase pipeline failure points.



The Unified Postgres Fix

- 1 Store vector embeddings directly alongside relational tables.
- 2 Perform hybrid search and standard SQL filters in one query.
- 3 Leverage existing backup, auth, and monitoring pipelines.

Clean Hybrid Search in SQL

- 1 Query relational metadata and vector similarity in a single step.

main.py

```
SELECT product_id, name, 1 - (embedding <=> $1) AS similarity
FROM products
WHERE category = 'electronics'
ORDER BY similarity DESC
LIMIT 10;
```

The Production Numbers

- 1 Runs at ~80% lower cost than specialized vector SaaS.
- 2 Easily supports up to 50M vectors with pgvectorscale indexing.



The Rule of Thumb

- 1 Default to Postgres + pgvector for simple, rock-solid architecture.
- 2 Only move to dedicated vector engines if you hit billion-scale loads.

